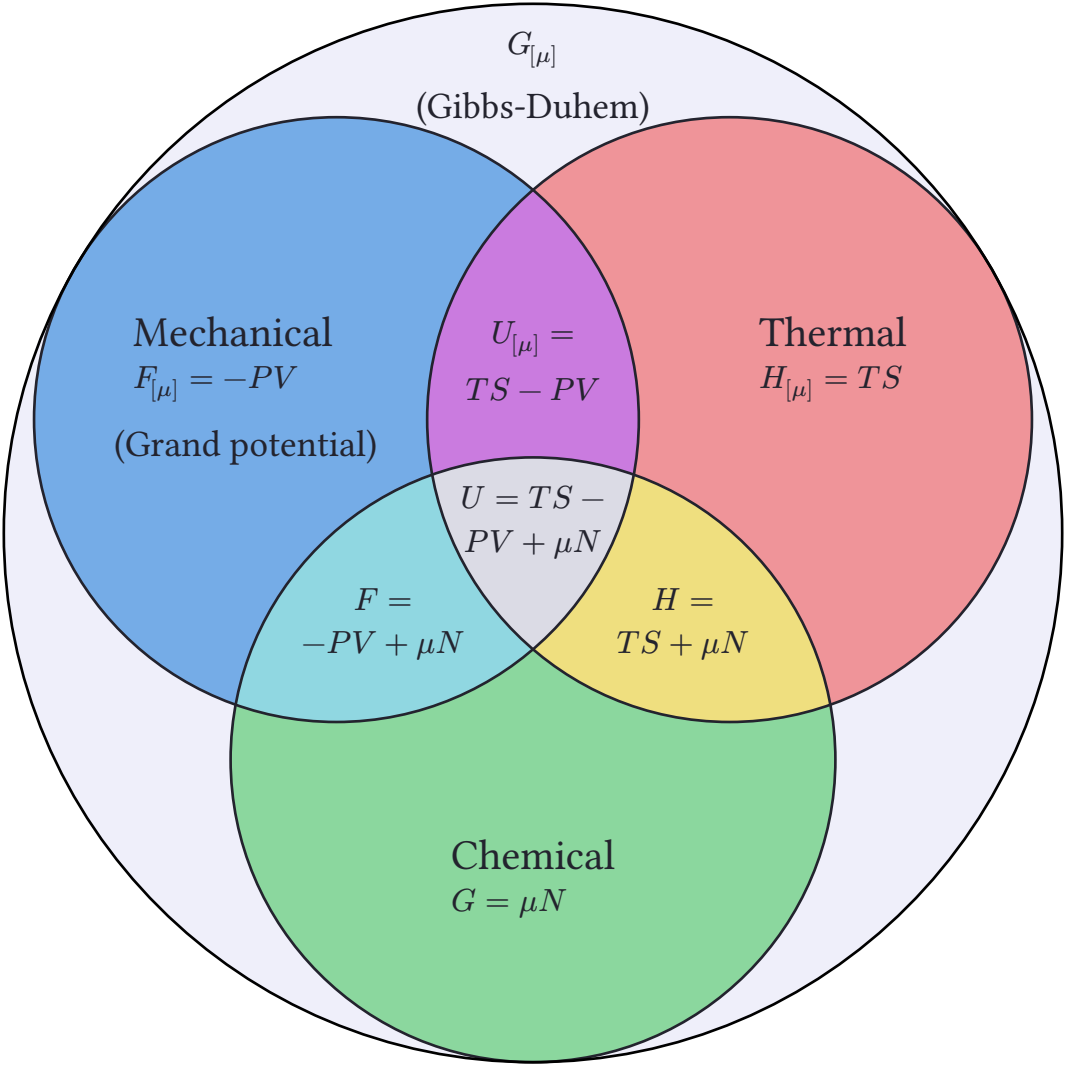
$G_{[\mu]}$
(Gibbs-Duhem)
Mechanical
$F_{[\mu]} = -PV$
(Grand potential)
$U_{[\mu]} = TS - PV$
Thermal
$H_{[\mu]} = TS$
$U = TS - PV + \mu N$
$F = -PV + \mu N$
$H = TS + \mu N$
Chemical
$G = \mu N$